EX NO: 4

Date: 10/08/26

PRACTICAL - 4 Setting Up a Local Area Network

AIM:

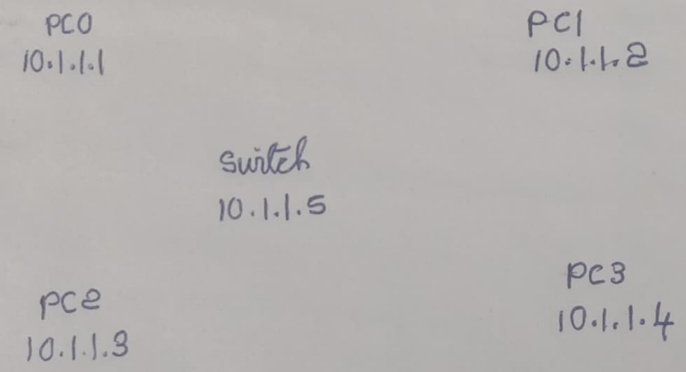
To design, set up and configure a Local Area Network connecting four Computer through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

Equipment Required:

S.NO	Equipment	Quantity	Specification
1	Personal Computer	4	ethernet (NIC) enabled
2	Network Switch	1	8/16/24 port, unmanaged/managed
3	Ethernet Cables (Cat5e/cat6)	4	Straight-through, RJ-45 connectors.

Network Diagram:

The LAN was implemented using a star Topology as shown below, with 4 PCs connected individually



IP Address Configuration Table:

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (Managed)	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation in Cisco Packet Tracer

Since physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer Simulation software, following the steps below.

Step 1: Place the Devices:

1. Open Cisco Packet Tracer
2. From the devices category panel, select End Devices and drag 4 PCs onto the workspace
3. Select Switches and drag 1 switch (Eg: 2960 Series) onto the workspace.

Step 2: Cable the Devices:

4. Click Connections → Select Copper Straight-Through Cable
5. Click each PC, then click the switch and select an available Fast Ethernet port to complete each connection
6. Repeat for all four PCs (Ports Fa0/1 to Fa0/4). Green link lights confirm a successful physical connection.

Step 3: Assign IP Addresses to PCs

7. Click a PC → Desktop tab → IP Configuration
8. Enter the static IP Address and Subnet Mask as per the IP table for each PC

Step 4: Configure the Switch Management IP

9. Click the Switch → CLI tab
10. Enter the following commands to assign a management IP

```
Enable
Configure Terminal
Interface vlan 1
ipaddress 10.1.1.5255.0.0.0
no shutdown
Exit
```

Step 5: Verify Connectivity Using Ping

11. Click a PC → Desktop tab → Command prompt
12. Use the ping Command to test Connectivity to the other PCs, eg: Ping 10.1.1.2
13. Successful "Reply from..." messages Confirm Connectivity

Step 6: Demonstrate Data Sharing (Server Simulation)

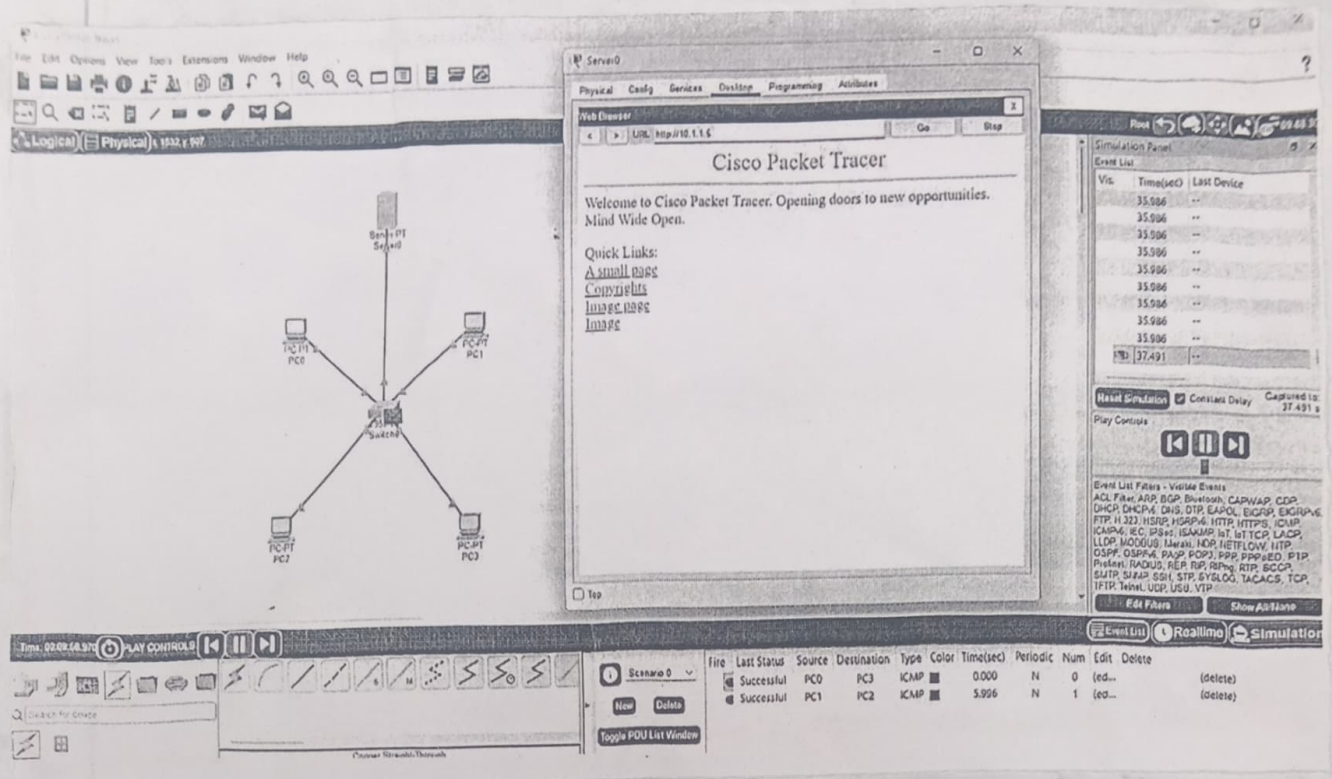
14. Add a Server device (End Devices → Server-PT) and connected it to the switch
15. Assign it an IP address (eg, 10.1.1.10, Subnet mask 25.0.0.0) via its Desktop → IP configuration tab
16. Open the Server's services (config) tab and enable the HTTP Service
17. From another PC, open the web Browser (Desktop tab) and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

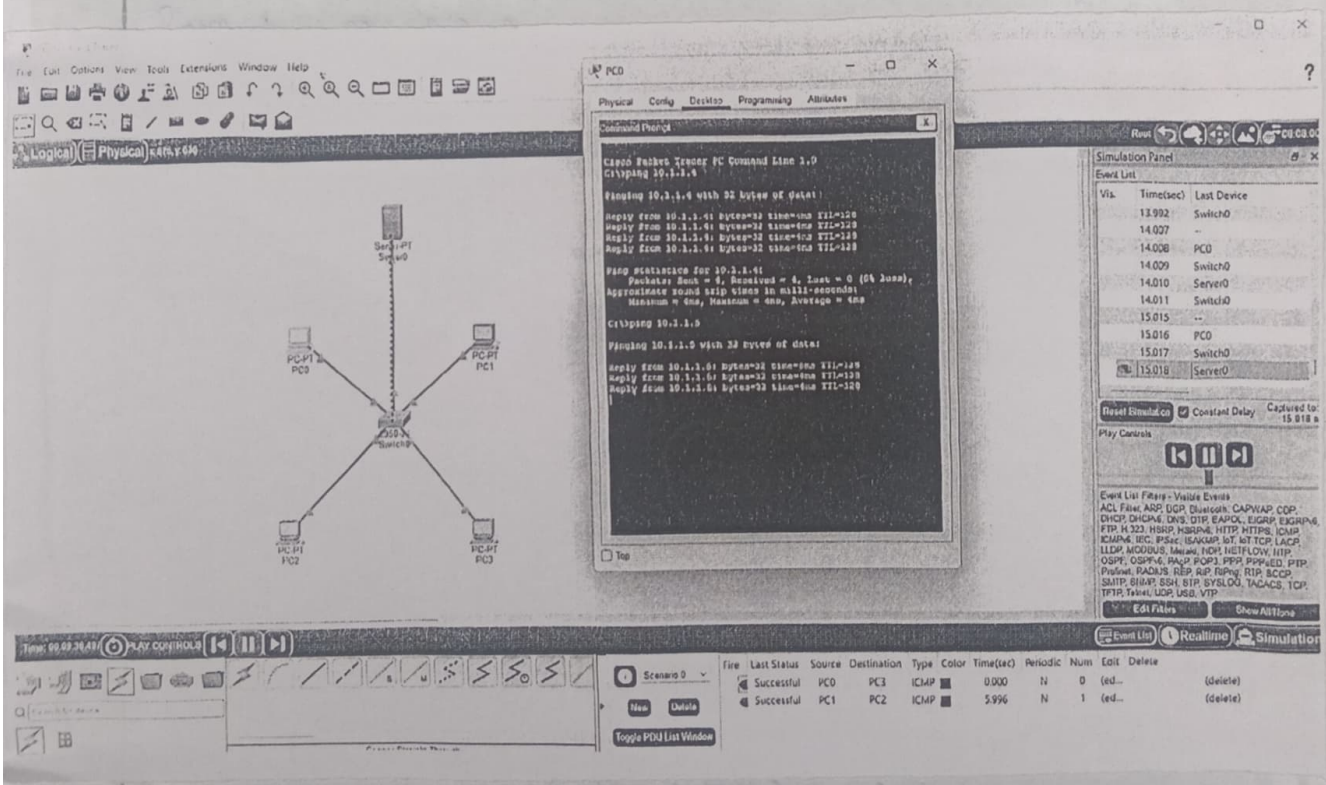
Step 7: Observe Packet Flow in Simulation Mode:

18. Switch to Simulation mode (bottom-right of the workspace)
19. Send a ping or web request between two devices and use the play/capture controls to step through the packet as it travels hop by hop through the switch, observing MAC address based forwarding

Observation

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP address within the same subnet Ping tests between all devices returned successful replies confirming end to end connectivity.





RESULT :

Thus setting up LAN was executed successfully